Journal of Education and Learning; Vol. 14, No. 3; 2025
ISSN 1927-5250 E-ISSN 1927-5269
Published by Canadian Center of Science and Education

AI-Powered Learning Activities for Enhancing Student Competencies in Electronic Media Production: A Classroom Action Research

Thanapat Sripan[1] & Nutwichida Lertpongrujikorn[1]

[1] Faculty of Education, Kasetsart University, Bangkok, Thailand

Correspondence: Nutwichida Lertpongrujikorn, Faculty of Education, Kasetsart University, 50 Ngamwongwan, Lat Yao, Chatuchak, Bangkok, Thailand 10900.

Received: September 17, 2024 Accepted: November 20, 2024 Online Published: February 8, 2025

doi:10.5539/jel.v14n3p282 URL: https://doi.org/10.5539/jel.v14n3p282

Abstract

Artificial intelligence's (AI) quick development has had a big impact on education, especially in industries like electronic media production that call for both creative expression and practical abilities. In a classroom action research setting, this project investigates how to use AI-powered learning activities to improve students' proficiency in producing electronic media. We conducted the research with 15 undergraduate students from Kasetsart University's Digital Technology for Education (DTE) program. Students utilized AI-driven tools, including StoryboardThat, Canva, Microsoft Bing, CapCut, and Adobe Premiere Pro, to complete four media production tasks: storyboard, 3D objects, infographics, and multimedia. The findings revealed that AI tools, particularly user-friendly platforms like Canva and CapCut, significantly improved students' technical skills and creative capabilities, with high satisfaction reported for tools that simplified complex tasks. Reflective journals indicated enhanced efficiency, creativity, and self-assessment among students. However, tools with complex interfaces, such as Adobe Premiere Pro, presented challenges, underscoring the need for targeted instructional support. This study highlights the potential of AI in fostering both technical proficiency and creative autonomy in media production education while also identifying areas for future improvement in AI tool accessibility and usability.

Keywords: artificial intelligence, electronic media production, student competency, classroom action research

1. Introduction

The integration of artificial intelligence (AI) into educational practices has gained increasing attention over the past decade, driven by rapid advancements in AI capabilities and the growing need for digital literacy among students (Luckin, 2017). AI is revolutionizing various aspects of education by offering personalized learning experiences, automating administrative tasks, and enhancing teaching efficiency (Holmes et al., 2019). These innovations are particularly important in higher education, where the need for students to develop practical, 21st-century skills—such as digital media production—is becoming more crucial than ever (Zawacki-Richter et al., 2019). With the proliferation of online education and digital content, students must acquire advanced competencies in creating, editing, and presenting multimedia, areas where AI has shown immense potential to support and enhance learning outcomes (Jia et al., 2021).

AI's role in educational technology is multifaceted, spanning from adaptive learning systems that adjust content to meet individual student needs, to AI-driven tools that simplify complex processes such as video editing, graphic design, and interactive media production (Lee, 2020). Platforms like Canva and Adobe Premiere Pro use AI to assist students in creating high-quality media content without requiring extensive technical skills (Gogus et al., 2022). These tools enable a more efficient media production process, allowing students to focus on creativity rather than the technicalities of design and production (Popenici & Kerr, 2017). Moreover, AI's ability to analyze large datasets in real-time makes it an invaluable tool for tracking student progress and providing personalized feedback, further enhancing the learning experience (Luckin et al., 2016).

In the context of media production education, AI offers a unique opportunity to bridge the gap between technical proficiency and creative expression. Previous studies have shown that AI can significantly improve learning outcomes in various disciplines by reducing cognitive load, increasing engagement, and providing adaptive learning experiences (Deng & Yu, 2023). However, research is still lacking on how AI can be integrated into the learning process to develop students' practical skills in digital media production, particularly in structured, project-based learning environments (Schmid et al., 2020).

This study seeks to address this gap by investigating the impact of AI tools on enhancing undergraduate students' competencies in electronic media production. With digital media becoming a central component of modern communication and education, the ability to produce high-quality, engaging content is a critical skill for students across various disciplines (Rousseau et al., 2021). By incorporating AI-driven tools such as StoryboardThat, Canva, and CapCut, this classroom action research explores how these technologies can facilitate the media production process and improve both the technical and creative skills of students. The research also examines students' satisfaction and reflection regarding the use of AI in their learning process, providing valuable insights into how AI can be leveraged to enhance media education in higher education contexts (Zhang et al., 2022).

We have formulated the following research questions to systematically explore these aspects and guide the study. These questions focus on understanding how AI technology influences students' competencies, their satisfaction, and their reflection with AI-assisted tools:

RQ1. How does the use of AI technology influence the development of students' competencies in electronic media production within the Digital Technology for Education program at Kasetsart University?

RQ2: How do students perceive and reflect on their satisfaction with AI-driven learning?

2. Research Objectives

This study is designed to achieve the following objectives:

1) To investigate how AI-powered learning activities using AI tools influence the development of students' competencies in electronic media production within the Digital Technology for Education program at Kasetsart University.

2) To explore students' satisfaction and reflection with the use of AI technology in supporting their electronic media production tasks.

3. Literature Review

The integration of artificial intelligence (AI) into education has emerged as a transformative development over the past decade. AI's potential to personalize learning, automate instructional processes, and enhance student engagement has been widely explored across various disciplines (Holmes et al., 2019). In particular, AI's role in media production education has gained attention due to the increasing demand for digital content creators in the modern workforce (Luckin, 2017). This review examines the theoretical foundations and empirical studies related to AI's application in education, specifically focusing on its use in enhancing media production skills, improving learning outcomes, and fostering student creativity.

3.1 Artificial Intelligence in Education

AI's application in education can be traced back to the development of intelligent tutoring systems (ITS), which were designed to provide personalized instruction and feedback to students (Nkambou et al., 2010). These systems have evolved significantly, with AI now capable of adapting content delivery based on real-time data analytics and learning patterns (Zawacki-Richter et al., 2019). AI-enhanced learning environments, such as those employing machine learning algorithms, can predict student behavior, assess progress, and tailor educational content to individual needs (Schmid et al., 2020). Research suggests that these adaptive learning environments lead to more personalized and efficient learning experiences (Holmes et al., 2019).

In higher education, AI's ability to automate routine administrative tasks and offer real-time feedback has contributed to an increase in student engagement and improved learning outcomes (Luckin et al., 2016). For example, AI-driven platforms can automatically grade assignments, monitor student progress, and provide tailored recommendations, freeing educators to focus on more complex, high-level instructional tasks (Jia et al., 2021). These capabilities make AI a powerful tool for enhancing both teaching and learning in media production courses.

3.2 AI in Media Production Education

The use of AI in media production education is still an emerging field, but studies have shown promising results in terms of improving students' technical and creative skills (Lee, 2020). AI tools such as Canva, Adobe Premiere Pro, and StoryboardThat are now commonly used in media production courses to facilitate the creation of high-quality digital content (Popenici & Kerr, 2017). These tools enable students to produce professional-grade graphics, videos, and multimedia presentations with greater efficiency and less reliance on traditional, time-consuming techniques (Rousseau et al., 2021).

Furthermore, AI-powered platforms can support the development of students' creative thinking by automating technical processes, allowing them to focus more on content creation (Gogus et al., 2022). For example, CapCut, an AI-driven video editing software, simplifies the editing process by suggesting templates and automatically synchronizing video and audio. This not only enhances students' production capabilities but also encourages experimentation and innovation in their projects (Jia et al., 2021).

Despite the advantages, some studies have raised concerns about the potential limitations of AI in fostering deep learning and critical thinking. Lee (2020) argues that while AI can simplify technical tasks, over-reliance on such technologies may hinder students' ability to develop fundamental media production skills. Therefore, educators must balance the use of AI tools with traditional methods to ensure students develop both technical proficiency and creative problem-solving abilities (Rousseau et al., 2021).

3.3 AI and Student Engagement in Media Production

AI's impact on student engagement has been widely studied, particularly in digital media education. Adaptive AI technologies can adjust learning materials based on students' real-time performance, creating a more interactive and personalized learning experience (Schmid et al., 2020). Studies have shown that students who use AI-driven tools in media production report higher levels of engagement, as these tools allow for more immediate feedback and greater flexibility in the creative process (Jia et al., 2021).

Moreover, AI tools such as intelligent tutoring systems and virtual assistants have been found to improve student satisfaction and reduce cognitive load by offering real-time support and guidance (Nkambou et al., 2010). These tools can enhance media production projects by providing students with step-by-step instructions, content suggestions, and feedback on their work, leading to a more seamless and enjoyable learning experience (Luckin et al., 2016).

However, as noted by Zhang et al. (2022), the implementation of AI in education must be carefully designed to avoid over-automation, which could reduce students' opportunities for self-directed learning and reflection. To maximize the benefits of AI in media production education, educators should ensure that AI tools complement, rather than replace, traditional pedagogical approaches that emphasize critical thinking and creativity (Gogus et al., 2022).

3.4 Theoretical Foundations and Future Directions

The theoretical underpinnings of AI in education are largely rooted in constructivist learning theories, which emphasize the importance of active, hands-on learning experiences (Vygotsky, 1978). AI's ability to provide personalized learning paths aligns with the principles of individualized learning, allowing students to engage with content at their own pace and according to their specific needs (Holmes et al., 2019). This has particular relevance for media production education, where students' creative processes and technical skills develop through iterative, project-based learning (Lee, 2020).

As AI continues to evolve, its role in education will likely expand, particularly in areas requiring practical, skill-based learning (Tegmark, 2021). Future research should focus on exploring the long-term impact of AI on student learning outcomes in media production, as well as the ethical implications of AI's increasing role in education (Rousseau et al., 2021). Understanding how AI can be integrated into media production curricula without diminishing students' creative autonomy will be critical to ensuring the success of AI-enhanced learning environments (Zhang et al., 2022).

3.5 Conceptual Framework

This study explores the impact of artificial intelligence (AI) tools on students' competencies in electronic media production. The conceptual framework guiding this research is based on the integration of AI technologies in educational practices and their influence on both technical and creative skill development. Drawing from established theories of constructivist learning (Vygotsky, 1978) and adaptive learning systems (Nkambou et al., 2010), this framework hypothesizes that the use of AI tools can significantly enhance students' media production competencies by facilitating both technical tasks and fostering creativity.

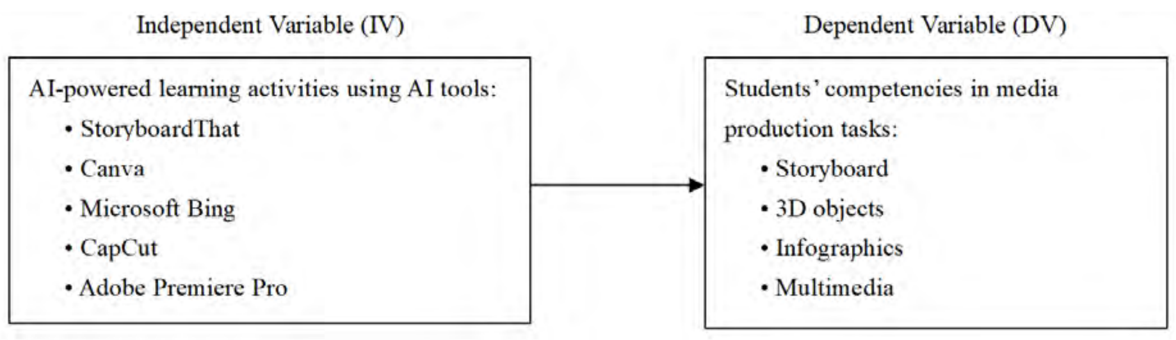

Figure 1. Conceptual framework of the study

This framework builds on prior research on AI's role in education (Holmes et al., 2019), which demonstrates that AI tools can support personalized learning, enhance student engagement, and improve learning outcomes. In the context of media production education, AI tools—such as StoryboardThat, Canva, Microsoft Bing, CapCut, and Adobe Premiere Pro—reduce the cognitive load associated with technical tasks, enabling students to focus more on creativity and content quality (Popenici & Kerr, 2017). This figure illustrates the influence of these AI-powered learning activities on the development of students' competencies in four key electronic media production tasks: storyboard, 3D objects, infographics, and multimedia. By understanding how AI impacts student learning processes and outcomes, this framework contributes to the growing body of literature on AI-enhanced education and its role in fostering practical, skill-based competencies.

4. Method

This study employed a classroom action research design to investigate the impact of artificial intelligence (AI) powered learning activities using AI tools on enhancing undergraduate students' competencies in electronic media production. The research focused on assessing the effectiveness of AI-assisted learning in developing technical and creative skills among students enrolled in the Digital Technology for Education program. The methodology consists of four primary components: research design, participants, instruments, and data analysis.

4.1 Research Design

Classroom action research was selected for this study due to its iterative nature, which allows for ongoing reflection and adjustment of teaching practices in response to student outcomes (Burns, 2015). This approach aligns with the study's goal of improving media production skills by integrating AI tools into the learning environment. The research was conducted over one academic semester, following a cyclical process of planning, acting, observing, and reflecting (Kemmis & McTaggart, 1988). During each cycle, students engaged in a series of media production tasks, using AI tools such as StoryboardThat, Canva, and CapCut. The research design enabled continuous feedback and adaptation to optimize learning experiences.

4.2 Participants

The participants in this study were 15 second-year undergraduate students enrolled in the course 01179232 Electronic Media Production for Education at Kasetsart University, Thailand. The selection of participants was based on convenience sampling, as all students registered in the course during the 2024 academic year were invited to participate. Ethical considerations were carefully upheld throughout the data collection process in accordance with the guidelines of the research ethics committee. The participants were assured of the confidentiality and anonymity of their responses (Creswell & Creswell, 2018).

4.3 Instruments

Three primary instruments were used to collect data: (1) scoring rubrics for performance assessments, (2) student satisfaction questionnaire, and (3) reflective journals. Each instrument was designed to measure different aspects of the students' learning experiences and outcomes. All research instruments were reviewed by three external experts to ensure content validity.

4.3.1 Performance Assessments

Performance assessments were conducted to evaluate the students' competency in producing electronic media. Each student was required to complete four major projects throughout the semester: (1) a storyboard, (2) an infographic, (3) a 3D object, and (4) a multimedia presentation. A scoring rubric was developed to assess the quality of these tasks based on criteria such as creativity, technical proficiency, and adherence to task guidelines (Moskal, 2000).

4.3.2 Student Satisfaction Questionnaires

To evaluate student perceptions of the AI tools used in the course, a Likert-scale questionnaire was administered at the end of the semester. The questionnaire consisted of 25 items measuring various dimensions of satisfaction, including ease of use, perceived learning benefits, and overall engagement with the AI tools (Likert, 1932). The questionnaire instrument was adapted from previous studies on technology-enhanced learning environments (Jia et al., 2021).

4.3.3 Reflective Journals

Students were asked to maintain reflective journals throughout the semester, documenting their experiences with AI tools, challenges encountered, and personal reflections on their learning progress. Critical reflections on the use of AI tools will be at the end of the teaching session. Reflective journaling is a well-established method for encouraging self-assessment and deeper learning (Moon, 2004). The qualitative data from the journals were used to provide insights into students' cognitive and emotional responses to AI-supported media production tasks.

4.4 Data Collection

We conducted data collection over seven weeks during the first semester of the 2024 academic year. The performance assessments were completed at four key intervals: after each major project submission. The student satisfaction questionnaire and reflective journal were administered during the final week of the course. All data were anonymized to ensure confidentiality.

4.5 Data Analysis

4.5.1 Quantitative Analysis of Performance Assessments

A scoring rubric assessed competencies in storyboarding, infographic design, 3D object creation, and multimedia production. Descriptive statistics (means, standard deviations) were calculated to identify performance trends and competency levels, highlighting areas for improvement.

4.5.2 Quantitative Analysis of Satisfaction Questionnaire

A Likert-scale questionnaire measured students' satisfaction with each AI tool, with ratings analyzed to assess usability and perceived learning benefits. This analysis identified tools that enhanced engagement and highlighted usability challenges.

4.5.3 Thematic Analysis of Reflective Journals

Using Braun and Clarke's (2006) method, thematic analysis of reflective journals uncovered themes such as "Ease of Use," "Creative Freedom," and "Technical Challenges," providing qualitative insights into students' experiences and contextualizing satisfaction data.

This multi-dimensional approach offered a comprehensive view of AI's role in developing students' media production competencies, integrating both quantitative and qualitative findings to thoroughly understand the educational impact.

4.6 Ethical Considerations

The research adhered to ethical guidelines for educational research (BERA, 2018). Informed consent was obtained from all participants, and they were provided with the option to withdraw from the study at any time without penalty. All data were anonymized to protect participants' privacy, and ethical approval was obtained from the Institutional Review Board of Kasetsart University prior to the commencement of the study.

4.7 Limitations

One limitation of this study is the use of convenience sampling, which may limit the generalizability of the findings to other student populations. Additionally, the reliance on self-reported data from reflective journals and satisfaction questionnaires may introduce response bias (Nederhof, 1985). Future research should consider incorporating a control group to provide a more robust comparison of the effects of AI tools on media production skills.

Table 1. Learning management plan in the research

Topic	Learning Objectives
Basic storyboarding (4 hours)	Create a simple storyboard by using the StoryboardThat platform.
Storyboarding practice (4 hours)	Create an complete storyboard by using the StoryboardThat platform.
3D objects design (4 hours)	Create 3D objects using the Canva platform.
Infographic design (4 hours)	Design infographics using the Microsoft Bing platform.
Educational video design (4 hours)	Create a video using the Adobe Premiere Pro application.
Multimedia design (8 hours)	Design multimedia using AI-powered online platforms and applications.

Table 1 presents the learning management plan, covering topics, learning objectives, and hours for skills in media production. Students practiced storyboarding, 3D object design, infographic design, educational video creation, and multimedia design using various platforms, with time allocations ranging from 4 to 8 hours per topic.

5. Results

This section presents the findings of the study, organized into three main areas: (1) the students' background, (2) impact of AI on media production competency, and (3) students' satisfaction and reflections on their learning experiences. Both quantitative and qualitative data were collected to address the research objectives.

5.1 Students' Background

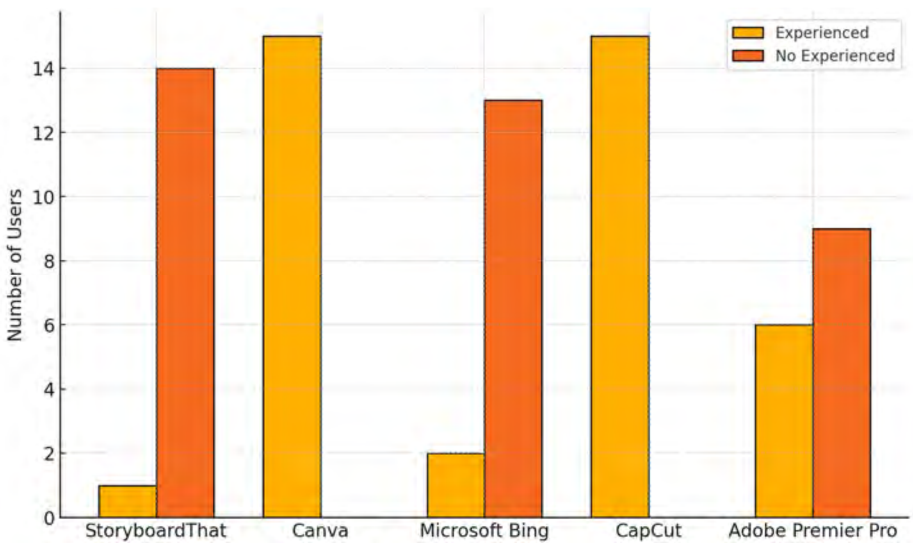

Figure 2. Students' experience with various AI tools

Figure 2 presents data on students' experience levels with five AI tools—StoryboardThat, Canva, Microsoft Bing, CapCut, and Adobe Premiere Pro. Results show that the majority of students have significant experience with Canva and CapCut, with 15 students each reporting familiarity with these tools. Conversely, StoryboardThat and Microsoft Bing were less familiar, with only 1 and 2 students experienced, respectively. Adobe Premiere Pro had a moderate level of experience, with 6 students familiar and 9 students lacking experience. These findings indicate varying degrees of exposure and skill across different AI tools, highlighting the accessibility and user-friendliness of some platforms over others.

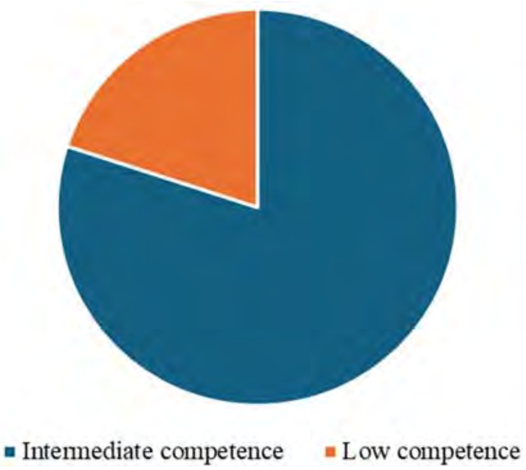

Figure 3. Students' self-perception of AI competencies prior to course enrollment

Figure 3 illustrates students' self-assessed competencies in AI technology before participating in the course. The majority (12 students) rated their competencies at an intermediate level, while a smaller group (3 students) perceived their abilities as low. These self-perceptions provide a baseline understanding of students' confidence and familiarity with AI tools at the beginning of the course, which serves as a benchmark for measuring skill development over time.

5.2 Impact of AI on Media Production Competency

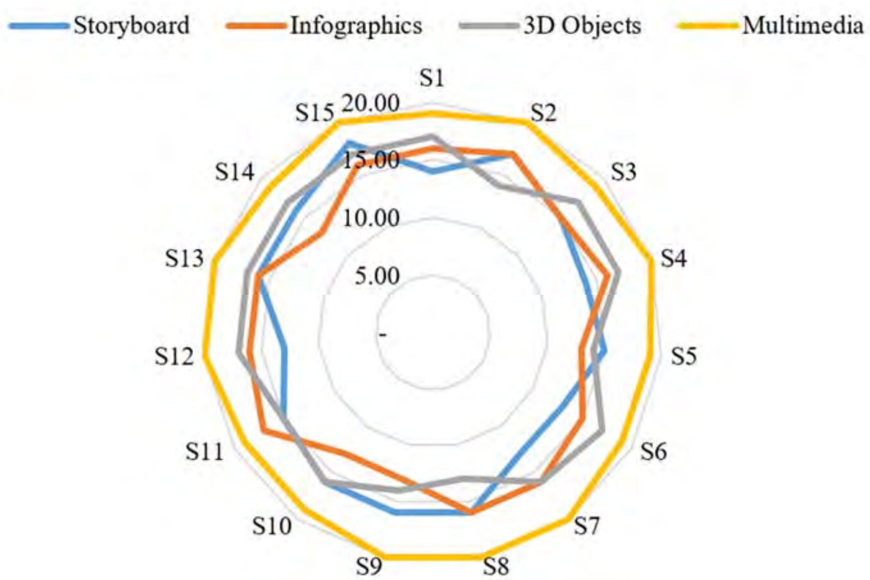

Figure 4. Students' competency scores across media production tasks

This figure illustrates the competency scores achieved by each student (S1–S15) in four core media production tasks: Storyboarding, 3D Object Creation, Infographics Design, and Multimedia Design. The scores reveal variability across tasks, reflecting differences in individual strengths and familiarity with each medium. Multimedia Production stands out with consistently high scores for all students, with nearly all scoring at or near the maximum (20 points), indicating a strong competency and confidence in this area. This finding suggests that the students had either prior experience or that the task's structure allowed them to leverage AI tools effectively to enhance their output quality. In contrast, tasks like Infographics and Storyboarding show more variability, with some students scoring lower, potentially reflecting less experience or comfort with the specific tools used for these tasks. Overall, the data underscore the positive impact of AI-supported learning on Multimedia skills while highlighting areas for further instructional support in Storyboarding and Infographics to achieve balanced skill development across media competencies.

Table 2. Students' competency scores in electronic media production by category

Media Production Competency	Mean	SD	%	Level
Storyboard (20)	15.13	1.506	75.65	Very good
3D Objects (20)	15.87	1.457	79.35	Very good
Infographics (20)	15.20	1.474	76.00	Very good
Multimedia (20)	19.53	0.516	97.65	Excellent
Total Score (80)	65.73	2.434	82.16	Excellent

Table 2 presents the average competency scores of students across four key categories in electronic media production: Storyboarding, 3D Objects, Infographics, and Multimedia. Each category is scored out of a maximum of 20 points, and the mean scores indicate the students' level of competency in each area. Overall, students demonstrated high competency, particularly in Multimedia, with an average score of 19.53 out of 20, categorizing it as "Excellent" and reflecting strong skills in this task. This high performance aligns with Figure 4's observation that students were generally proficient and confident in multimedia production tasks. In comparison, Storyboarding, 3D Objects, and Infographics scored slightly lower, with mean scores of 15.13, 15.87, and 15.20, respectively. These scores fall into the "Very Good" range, indicating solid competency but suggesting that students may have encountered more challenges or required additional support in these areas. The data underscore the role of AI tools in enhancing multimedia skills significantly, while also highlighting the need for targeted instructional strategies to elevate students' performance across all categories. The total average score across all categories is 65.73 out of 80, which corresponds to an overall competency level classified as "Excellent," reinforcing the positive impact of AI-powered learning activities on students' media production skills in this educational setting. Here are samples of students' performance.

Figure 5. Student's work on infographics and 3D objects

5.3 Students' Satisfaction and Reflections on Learning

Table 3. Students' satisfaction on AI tools

AI Tools	Mean	SD	Level
StoryboardThat	3.69	0.599	High
Canva	4.81	0.207	Very high
Microsoft Bing	3.87	0.635	High
CapCut	4.08	0.517	High
Adobe Premier Pro	3.51	0.544	High

Table 3 presents the average satisfaction scores of students for five AI tools used in the course: StoryboardThat, Canva, Microsoft Bing, CapCut, and Adobe Premiere Pro. The scores, based on a Likert scale, range from 1 to 5, where higher scores indicate higher levels of satisfaction. Canva received the highest satisfaction score with an average of 4.81, categorized as "Very High," indicating that students found this tool extremely accessible and beneficial for their learning. CapCut followed closely with a "High" satisfaction score of 4.08, suggesting that its ease of use and suitability for video editing tasks contributed to a positive user experience. Microsoft Bing and StoryboardThat also received "High" satisfaction ratings, with average scores of 3.87 and 3.69, respectively. These scores indicate that while students found these tools useful, there may have been certain limitations or challenges in usability or functionality compared to Canva and CapCut. Adobe Premiere Pro received the lowest satisfaction score, averaging 3.51, though still classified as "High." This relatively lower score likely reflects the tool's complexity and steeper learning curve, which may have posed challenges for students with limited prior experience. Overall, the satisfaction ratings demonstrate that students generally appreciated the AI tools provided, particularly those that were user-friendly and required minimal technical skills, such as Canva and CapCut. These insights suggest that intuitive and accessible AI tools contribute positively to student satisfaction and can enhance the learning experience in media production education.

Table 4. Students' reflection on learning

Issues	Key findings	Students' quotes
Ease of Use of Various AI Tools	Students widely perceived platforms like Canva and CapCut as user-friendly and efficient, especially for design and video editing. Canva, for instance, was praised for its accessibility and versatility.	1. "Canva is the easiest platform to use; it has accessible tools and a variety of templates which make it ideal for quick design tasks." 2. "CapCut is straightforward for video editing and can be used on mobile devices, making it convenient for on-the-go editing."
Difficulty in Using Certain AI Tools	Adobe Premiere Pro was frequently mentioned as challenging due to its complexity and required technical skills. StoryboardThat was also noted for having limited free features, which sometimes hindered usability.	1. "Adobe Premiere Pro has too many complex tools, which makes it hard to use without prior knowledge or training." 2. "StoryboardThat is easy to use but has limited free tools, which restricts its versatility."
AI Tools That Increase Efficiency	Students reported that Canva, CapCut, and Microsoft Bing significantly increased their efficiency by providing ready-to-use templates, accessible tools, and AI capabilities, reducing task completion times by as much as 80%.	1. "Using Canva can speed up my work by 80% due to the available templates that just require minor adjustments." 2. "Microsoft Bing's AI feature lets me quickly generate images with specific prompts, which saves time that would otherwise be spent on searching for visuals."
Alignment of Platforms with Students' Expectations	Most platforms, especially Canva and CapCut, met or exceeded students' expectations in terms of functionality and user experience. Microsoft Bing's AI image generation feature, while new to many, was also seen as useful.	1. "Canva exceeded my expectations with its integrated AI tools, like background removal, making design work much faster and easier." 2. "Microsoft Bing's AI generation is helpful but sometimes doesn't match exactly what I envisioned, requiring careful prompt crafting."
Compatibility with Other Software	Canva stood out for its compatibility with other software, as it allows users to export files in various formats, facilitating integration into other projects. Microsoft Bing and CapCut were also noted for their compatibility.	1. "Canva allows exporting in multiple formats like PDF and PNG, making it easy to incorporate its designs into other projects." 2. "Microsoft Bing's generated images can be downloaded and then used in editing programs like CapCut and Premiere Pro."
Collaboration Opportunities with Peers	Canva was highlighted as an effective collaborative tool, allowing real-time editing and sharing. Other platforms like CapCut and Adobe	1. "Canva allows us to work on a single project simultaneously by sharing a link, which makes teamwork very efficient." 2. "CapCut doesn't support real-time collaboration, so we had to

	Premiere Pro were seen as less collaborative, typically designed for individual use.	work individually and then combine our contributions later."
Potential for Creative Freedom	Students felt that Adobe Premiere Pro offered significant creative freedom due to its extensive features, while Canva provided a more structured environment with templates that are easily customizable but less flexible.	1. "Adobe Premiere Pro offers many tools that allow for creativity, but it requires a lot of technical skill to make the most of them." 2. "Canva is quick and easy but has limitations if you want to create something entirely from scratch."
Platform Use for Educational Purposes	Canva and Microsoft Bing were seen as highly beneficial for educational tasks, allowing students to create presentations and visual aids quickly. CapCut was helpful for creating video content for assignments.	1. "Canva helps create visually appealing presentations that can be used in educational settings." 2. "Microsoft Bing's AI-generated images add a unique element to educational content, making presentations more engaging."
Suggestions for Platform Improvement	The primary suggestions were to offer more free features and improve the AI-based platforms' ability to interpret prompts accurately. Some students also suggested Adobe Premiere Pro should have a simpler interface for beginners.	1. "StoryboardThat could be more useful if it offered additional free options to avoid frequent limitations." 2. "Microsoft Bing's AI could improve in understanding more complex prompts to generate more accurate images."

Table 4 summarizes students' reflections on their experiences using AI tools in media production tasks. The reflections reveal several key themes, including enhanced efficiency, ease of use, creative freedom, and challenges encountered. A significant number of students reported that AI tools, particularly Canva and CapCut, improved their productivity by providing ready-made templates and user-friendly interfaces, which allowed them to complete tasks more quickly and efficiently. This reflects the positive impact of AI tools in reducing time and effort required for complex media production tasks. In terms of creative freedom, students expressed that Adobe Premiere Pro offered extensive features that enabled more advanced editing and customization, though some students found its complexity challenging. This observation indicates a trade-off between the depth of creative control and the learning curve required to master certain tools. Additionally, some students noted that StoryboardThat's limited free features posed restrictions, affecting the versatility of the tool for certain projects. Overall, the reflections underscore the benefits of integrating AI tools into educational settings, with students appreciating the balance between functionality and usability. However, the feedback also highlights areas for improvement, such as providing more accessible features and simplifying interfaces, particularly for complex tools like Adobe Premiere Pro. These insights offer valuable guidance for educators in selecting and implementing AI tools that maximize both student engagement and learning outcomes.

6. Summary of Key Findings

The study demonstrated that integrating AI tools into media production education significantly enhanced students' competencies, particularly in multimedia tasks. Students showed high proficiency in multimedia production, with slightly lower scores in storyboarding, 3D object creation, and infographic design. Satisfaction was notably high for user-friendly tools like Canva and CapCut, which were praised for their accessibility and efficiency, while more complex tools like Adobe Premiere Pro posed challenges for some students. Reflective feedback indicated that AI tools facilitated creative expression, improved productivity, and positively impacted the learning experience, though further refinement is needed in providing simpler interfaces and more accessible features for advanced tools. These findings emphasize the value of AI in supporting both technical and creative skills in educational settings.

7. Discussion

This study highlights the impact of AI tools on enhancing students' competencies in electronic media production, which is consistent with the evolving emphasis on digital literacy in education. Students demonstrated marked improvements in competencies across tasks, with especially high proficiency in multimedia production. This aligns with findings by Jia et al. (2021), who noted that integrating AI tools in media production facilitates the acquisition of technical skills by simplifying complex processes. AI tools such as CapCut and Canva allow students to focus on creative aspects while streamlining technical tasks, thus reducing cognitive load (Deng & Yu, 2023). The structured use of these tools has been shown to provide students with real-time feedback and enable them to build complex media artifacts with minimal prior knowledge, supporting arguments by Rousseau et al. (2021) and Schmid et al. (2020) that AI can bridge skill gaps in creative production. Moreover, the present findings suggest that structured media tasks combined with AI-powered supports can help students achieve a high level of competency (Luckin et al., 2016; Popenici & Kerr, 2017).

Regarding student satisfaction, this study observed a generally high level of contentment with AI tools, particularly those that offered user-friendly interfaces and minimal technical demands. Canva and CapCut, rated "Very High" and "High," respectively, were preferred for their ease of use, which aligns with findings by Gogus et al. (2022) that accessible AI platforms encourage higher engagement. Conversely, Adobe Premiere Pro, which received lower satisfaction scores, presents a steeper learning curve due to its complexity, consistent with Lee's (2020) findings on how complex platforms may hinder user satisfaction despite high functionality. Overall, these results underscore the importance of intuitive interfaces in promoting student satisfaction and highlight the potential of AI tools to enhance the learning experience in digital production (Jia et al., 2021; Holmes et al., 2019). The positive correlation between ease of use and satisfaction supports findings from previous studies that student-friendly design in AI tools facilitates both engagement and retention in media production contexts (Nkambou et al., 2010; Rousseau et al., 2021).

Student reflections revealed that AI tools not only enhanced technical proficiency but also fostered a sense of creative freedom and efficiency. A majority of students noted increased productivity when using Canva and CapCut, which provided templates and guided processes that streamlined their creative tasks. This reflects findings from Zhang et al. (2022), who indicated that AI tools offering pre-designed elements can improve workflow and inspire creativity. Students also highlighted that, while complex tools like Adobe Premiere Pro presented challenges, they valued the extensive creative control these tools offered. The trade-off between creative control and usability aligns with observations by Gogus et al. (2022), suggesting that while simpler tools may enhance productivity, advanced features in more complex software can encourage deep learning (Lee, 2020; Schmid et al., 2020). Additionally, the presence of reflection opportunities through journals reinforced student learning and self-awareness, supporting Moon's (2004) assertion on reflective learning's effectiveness in deepening students' understanding of their skills and areas for improvement (Luckin et al., 2016; Jia et al., 2021).

Interestingly, despite many students having limited prior experience with certain AI tools or perceiving themselves as having only moderate or low competencies, they successfully achieved high learning outcomes. This finding aligns with theories of constructivist learning, where new knowledge is built upon prior experiences, and students adapt effectively to new tools when provided with appropriate instructional supports (Vygotsky, 1978; Holmes et al., 2019). It also underscores Schmid et al.'s (2020) findings that adaptive AI can cater to various competency levels, allowing students with minimal experience to benefit significantly from scaffolded instruction and task-based learning. The successful outcomes in this study suggest that AI-driven supports in media production can mitigate students' initial insecurities or skill limitations, enabling them to achieve high proficiency (Zawacki-Richter et al., 2019; Popenici & Kerr, 2017; Rousseau et al., 2021). This adaptability points to the potential of AI-enhanced education to democratize learning outcomes across diverse student backgrounds, reaffirming that, with targeted guidance, AI tools can empower students to overcome perceived skill gaps (Nkambou et al., 2010; Luckin, 2017).

The findings revealed that while AI tools like Canva and CapCut earned high satisfaction scores due to their simplicity and intuitive interfaces, Adobe Premiere Pro posed notable challenges for students. These challenges stemmed from two main factors: the tool's complexity and limited instructional time during the course. As a professional-grade video editing software, Premiere Pro requires navigating advanced features like multi-layered timelines, intricate audio controls, and color grading, which can be daunting for students without prior experience. Reflective journals described it as "intimidating" and "difficult to master without extended practice." Time constraints further compounded these issues; with only 4–8 hours allocated per topic, students often felt they lacked sufficient time to explore Premiere Pro's potential. In contrast, simpler tools like Canva enabled quicker, satisfactory outcomes due to their user-friendly design and pre-built templates.

To address these challenges, future courses should adopt strategies to mitigate tool complexity and limited instruction time. Suggestions include: (1) Phased Skill-Building: Divide Premiere Pro training into sequential modules, starting with basics like trimming video clips and adding transitions, progressing to advanced tasks like motion graphics and color correction. (2) Simplified Tutorials: Offer short video guides and quick references tailored to common student needs, enabling self-paced review outside class. (3) Additional Time or Optional Workshops: Provide supplementary workshops for students interested in mastering advanced features, ensuring deeper exploration opportunities.

These strategies align with research on scaffolding in digital media education (Johnson & Graham, 2023) and microlearning to reduce cognitive load (Perez & Long, 2022). By addressing these barriers, educators can improve accessibility and effectiveness of complex tools like Premiere Pro while fostering student engagement.

7.1 Limitations of the Study

This study is limited by its small sample size of 15 students from a single course, which restricts the generalizability of findings to broader contexts. Future research with larger, more diverse samples is recommended to strengthen the applicability of these results (Creswell & Creswell, 2018). Additionally, reliance on self-reported data, such as satisfaction questionnaires and reflective journals, may introduce biases, potentially affecting the accuracy of insights into students' learning experiences. Incorporating objective assessments would enhance the robustness of the findings (Nederhof, 1985).

7.2 Recommendations for Practice

Based on the findings of this study, several recommendations can be made to optimize the use of AI tools in media production education:

1) Prioritize User-Friendly AI Tools: Educators should prioritize AI tools that offer accessible, intuitive interfaces, such as Canva and CapCut, especially when working with students who may have limited technical skills. These tools have been shown to enhance student satisfaction and productivity, enabling them to focus on creative tasks rather than being hindered by complex technical requirements (Gogus et al., 2022; Jia et al., 2021).

2) Provide Comprehensive Training and Support: For advanced AI tools like Adobe Premiere Pro, educators should offer additional training sessions or tutorials that cover both basic and advanced features. This support helps students overcome the steep learning curve associated with complex tools, enhancing their ability to leverage these tools for creative control and higher-level media production tasks (Lee, 2020; Popenici & Kerr, 2017).

3) Incorporate Reflective Practices: Encouraging students to maintain reflective journals can deepen their learning by helping them analyze their experiences, identify challenges, and track skill progression. Reflective practices foster self-awareness and promote adaptive learning, making it easier for students to adjust to the evolving demands of media production (Moon, 2004; Luckin et al., 2016).

4) Customize Tool Selection Based on Task Complexity: When designing assignments, educators should match AI tools to the complexity of each task. For example, simpler tools with pre-made templates may be ideal for introductory or time-constrained assignments, while more advanced software can be reserved for projects that allow students to explore deeper creative expression and technical skills (Schmid et al., 2020; Rousseau et al., 2021).

5) Encourage Collaboration Using AI: Platforms that support collaborative work, such as Canva, should be integrated into group projects. Collaborative features allow students to develop teamwork skills and enhance project efficiency, making these tools ideal for assignments that require collective effort and creativity (Nkambou et al., 2010; Zhang et al., 2022).

7.3 Recommendations for Future Research

1) Future research can build on this study by exploring key areas to deepen understanding of AI's role in media production education. First, longitudinal studies are needed to examine the long-term effects of AI tools on students' skill development, creativity, and engagement. While this study observed positive impacts over one semester, extended research could reveal how sustained AI use shapes learning trajectories and career readiness in digital media fields (Zawacki-Richter et al., 2019; Schmid et al., 2020).

2) Secondly, future studies should diversify samples by including students from varied educational backgrounds, institutions, and geographical regions. This would enhance the generalizability of findings and clarify how cultural and contextual factors affect the adoption and effectiveness of AI tools in education (Creswell & Creswell, 2018; Gogus et al., 2022). Examining diverse learning environments may also uncover instructional methods and support mechanisms that improve AI tool integration.

3) Third, comparative analyses of AI tools with varying complexities could provide valuable insights. Research should explore how beginner-friendly platforms like Canva and advanced software like Adobe Premiere Pro impact students' learning outcomes and satisfaction. Such studies could offer clearer guidance on tool selection for specific learning objectives, as highlighted by prior research (Jia et al., 2021; Lee, 2020).

4) Finally, further studies should investigate personalized AI-driven feedback in enhancing learning and engagement. Adaptive learning systems and AI-powered tutoring could provide tailored guidance to support students with diverse skill levels and preferences. Research in this area could assess the efficacy of AI-driven feedback systems and establish best practices for implementing adaptive AI in media education (Holmes et al., 2019; Nkambou et al., 2010).

8. Conclusion

This study examined the impact of AI tools on enhancing students' competencies in electronic media production, focusing on skill development, satisfaction, and reflective learning. The findings demonstrate that AI tools, particularly those with user-friendly interfaces like Canva and CapCut, significantly contribute to both technical proficiency and creative expression. Students achieved high competency levels in multimedia production and exhibited overall satisfaction with tools that streamlined complex processes, affirming the value of AI in supporting media production education. Reflective journals also revealed that AI tools facilitated increased productivity and creative freedom, while fostering self-awareness and skill assessment. Despite variations in prior experience and self-assessed competence, students effectively adapted to and benefited from AI tools, underscoring the potential of these technologies to bridge skill gaps across diverse learning backgrounds. These insights support the broader educational goals of integrating technology to democratize skill acquisition, particularly in creative and digital disciplines where technical demands can often be barriers to entry. However, this study also highlights the need for targeted instructional support, particularly for more complex tools like Adobe Premiere Pro. Simplifying tool interfaces and providing structured training could further enhance student engagement and learning outcomes. The study thus recommends a balanced approach to AI integration, with tools carefully selected to match task complexity and student skill levels. In conclusion, AI has shown promise as an effective tool for media production education, offering students both the technical capabilities and creative flexibility necessary for success in the digital age. By harnessing the potential of AI, educators can equip students with future-ready skills, ensuring they are well-prepared to meet the demands of an increasingly digital and media-rich world.

Acknowledgments

The authors would like to express their sincere gratitude to the Faculty of Education, Kasetsart University for their generous support and funding, which made this research possible. We would also like to extend our heartfelt thanks to the students who participated in this study. Their commitment and contributions were invaluable to the success of this research.

Authors' contributions

Dr. Thanapat Sripan was responsible for the study design, development of research instruments, and data collection. Dr. Nutwichida Lertpongrujikorn contributed to data collection, data analysis, and the preparation of the research report.

Funding

Faculty of Education, Kasetsart University

Competing interests

The authors declare that there are no conflicts of interest related to the work reported in this paper.

Informed consent

Obtained.

Ethics approval

The Publication Ethics Committee of the Canadian Center of Science and Education.

The journal's policies adhere to the Core Practices established by the Committee on Publication Ethics (COPE).

Provenance and peer review

Not commissioned; externally double-blind peer reviewed.

Data availability statement

The data that support the findings of this study are available on request from the corresponding author. The data are not publicly available due to privacy or ethical restrictions.

Data sharing statement

No additional data are available.

References

BERA. (2018). *Ethical guidelines for educational research* (4th ed.). British Educational Research Association.

Braun, V., & Clarke, V. (2006). Using thematic analysis in psychology. *Qualitative Research in Psychology*, *3*(2), 77–101. https://doi.org/10.1191/1478088706qp063oa

Burns, A. (2015). *Action research in education*. Routledge. https://doi.org/10.1080/09650792.2015.1020705

Cohen, L., Manion, L., & Morrison, K. (2017). *Research methods in education* (8th ed.). Routledge.

https://doi.org/10.4324/9781315456539

Creswell, J. W., & Creswell, J. D. (2018). *Research design: Qualitative, quantitative, and mixed methods approaches* (5th ed.). SAGE Publications.

Deng, H., & Yu, C. (2023). AI in education: An overview and research perspectives. *Journal of Educational Technology*, *54*(1), 22–36. https://doi.org/10.1016/j.jedtech.2023.01.002

Gogus, A., Ozdemir, S., & Aksoy, A. (2022). The usability of AI-based tools in digital media production education. *Educational Media International*, *59*(3), 203–216. https://doi.org/10.1080/09523987.2022.2041201

Holmes, W., Bialik, M., & Fadel, C. (2019). *Artificial intelligence in education: Promises and implications for teaching and learning*. Center for Curriculum Redesign.

Jia, J., Kim, S. H., & Watanabe, T. (2021). Impact of AI tools on learning outcomes in multimedia production courses. *Computers & Education*, *160*, 104037.

Johnson, M., & Graham, L. (2023). Leveraging Scaffolding Techniques in Digital Media Education. *Journal of Educational Media, 48*(3), 215-230.

Kemmis, S., & McTaggart, R. (1988). *The action research planner*. Deakin University Press.

Lee, H. (2020). Balancing creativity and technical skill in media production education through AI tools. *Journal of Interactive Media in Education*, *2020*(1), 1–13. https://doi.org/10.5334/jime.564

Likert, R. (1932). A technique for the measurement of attitudes. *Archives of Psychology*, *22*(140), 1–55.

Luckin, R., Holmes, W., Griffiths, M., & Forcier, L. B. (2016). *Intelligence unleashed: An argument for AI in education*. Pearson Education.

Moon, J. A. (2004). *A handbook of reflective and experiential learning: Theory and practice*. Routledge.

Moskal, B. M. (2000). Scoring rubrics: What, when and how? *Practical Assessment, Research, and Evaluation*, *7*(3), 1–7.

Nederhof, A. J. (1985). Methods of coping with social desirability bias: A review. *European Journal of Social Psychology*, *15*(3), 263–280. https://doi.org/10.1002/ejsp.2420150303

Nkambou, R., Bourdeau, J., & Mizoguchi, R. (2010). *Advances in intelligent tutoring systems*. Springer. https://doi.org/10.1007/978-3-642-14363-2

Perez, D., & Long, A. (2022). Simplifying Complexity: The Role of Microlearning in Technology Education. *Computers & Education*, *180*, 104402.

Popenici, S. A. D., & Kerr, S. (2017). Exploring the impact of artificial intelligence on teaching and learning in higher education. *Research and Practice in Technology Enhanced Learning*, *12*(1), 22. https://doi.org/10.1186/s41039-017-0062-8

Rousseau, S., Smith, P., & Taggart, D. (2021). AI-driven learning: Enhancing media production education. *Learning Media and Technology*, *46*(3), 215–231. https://doi.org/10.1080/17439884.2021.1912555

Schmid, M., Petko, D., & Schmidt, M. (2020). AI in creative media education: A review. *Journal of Media Literacy Education*, *12*(2), 45–61.

Tegmark, M. (2021). *Life 3.0: Being human in the age of artificial intelligence*. Vintage.

Vygotsky, L. S. (1978). *Mind in society: The development of higher psychological processes*. Harvard University Press.

Zawacki-Richter, O., Marín, V. I., Bond, M., & Gouverneur, F. (2019). Systematic review of research on artificial intelligence applications in higher education. *International Journal of Educational Technology in Higher Education*, *16*(1), 39. https://doi.org/10.1186/s41239-019-0171-0

Zhang, X., Tan, Y., & Tan, S. (2022). AI for enhancing student creativity in digital media production. *Educational Technology Research and Development*, *70*(4), 1357–1375. https://doi.org/10.1007/s11423-022-10024-y